Complete Project Audit

Knowledge Graph Analysis • Architecture Report • Code Quality Review

Generated: May 31, 2026

1,098	5,912	7,266	611	222x
Files Audited	Graph Nodes	Edges	Communities	Token Reduction

1. Executive Summary

This report presents a complete static analysis of the RoughTools.com monorepo, processed as a knowledge graph with 5,912 nodes and 7,266 edges across 611 detected communities. The analysis covers 1,098 source files — 1,063 code files parsed via AST and 35 documentation/PDF/image files via semantic extraction. The graph provides a 222x token reduction for future codebase queries.

Total files	1,098 (1,063 code · 29 docs · 4 PDFs · 2 images)
Graph nodes	5,912
Graph edges	7,266 (100% EXTRACTED via AST)
INFERRED edges	31 (avg confidence 0.83 — cross-file call relationships)
Communities	611 total · 124 thin (<3 nodes) omitted from detail
God nodes	10 high-betweenness cross-community bridges
Surprising connections	5 hidden cross-module dependencies
Knowledge gaps	2,925 isolated nodes (mostly package.json fields)
Token reduction	222x — graph query vs. reading full corpus
Date	May 31, 2026

2. God Nodes — Core Abstractions

God nodes are the highest-betweenness nodes in the graph — components that bridge multiple communities. A bug or breaking change in any of these nodes propagates across the widest surface area. Treat these as your highest-risk change points.

#	Node	Edges	Role
1	`JsonLd()`	73	SEO JSON-LD injected into every tool page
2	`getCalcSEO()`	50	SEO config factory for all calculator pages
3	`exportCalcPDF()`	27	PDF export shared across math/science calculators
4	`Card()`	25	Core UI card primitive — calculators, tools, landing
5	`ResultRow()`	25	Shared result row across calculator family
6	`scripts`	24	Build/test/lint script definitions
7	`InputField()`	22	Core form input used by all tool forms
8	`ApiError`	21	Central error type across Express + clients
9	`validateFile()`	19	File validation gate for all PDF uploads
10	`cn()`	19	Tailwind class merger — everywhere in UI

Critical observation: JsonLd() has 73 edges — the highest betweenness centrality in the entire graph (0.038). It bridges 60+ tool page communities to the SEO layer. A regression in this single component would break structured data on every tool page simultaneously, directly impacting Google indexing.

3. Surprising Connections

These cross-module dependencies were detected by the graph but are not obvious from reading any single file. They represent hidden coupling that could cause unexpected breakage during refactoring.

From	Relation	To	File Path
`Page()`	calls	`useClipboard()`	apps/web/src/app/api-docs/page.tsx → packages/ui/src/hooks/use-clipboard.ts
`Page()`	calls	`useClipboard()`	apps/web/src/app/embed-generator/page.tsx → packages/ui/src/hooks/use-clipboard.ts
`Page()`	calls	`useClipboard()`	apps/web/src/app/link-to-us/page.tsx → packages/ui/src/hooks/use-clipboard.ts
`SiteHeader()`	calls	`useTheme()`	apps/web/src/components/SiteHeader.tsx → packages/ui/src/providers/theme.tsx
`CI Job: E2E Tests (Playwright)`	shares_data_with	`PostgreSQL 16 Service`	.github/workflows/ci.yml → docker-compose.yml

4. Group Relationships (Hyperedges)

Hyperedges capture relationships among 3+ nodes that cannot be expressed as pairwise edges. These represent architectural clusters — stacks, pipelines, and feature groups that operate as a unit.

Group	Members	Conf.
Production Docker Services	docker_compose_production, docker_service_web, docker_service_db, docker_service_redis, docker_service_nginx	1.00
Development Docker Services	docker_compose_dev, docker_service_web, docker_service_db, docker_service_redis, docker_service_adminer	1.00
pnpm Workspace Packages	workspace_apps_web, workspace_pkg_calculators_core, workspace_pkg_db, workspace_pkg_cache, workspace_pkg_ui, workspace_pkg_storage	1.00
RoughTools 15 Tool Categories	project_category_pdf, project_category_calculators, project_category_medical, project_category_engineering, project_category_image_tools, project_category_developer, project_catego…	1.00
CI/CD Pipeline Stack	ci_pipeline, deploy_production_workflow, deploy_staging_workflow, deploy_generic_workflow	1.00
Monitoring & Analytics Stack	project_analytics_ga4, project_analytics_clarity, project_analytics_webvitals, project_analytics_adsense, project_sentry_middleware, project_api_logger, project_uptimerobot	1.00
Toolverse Database Backup Pipeline	backupcron_daily_incremental_backup, backupcron_weekly_full_backup, backupcron_backup_verification, backupcron_toolverse_db, backupcron_s3_backup_storage	1.00
Authentication API Endpoints	phase1security_api_auth_register, phase1security_api_auth_login, phase1security_api_auth_logout, phase1security_api_auth_refresh	1.00
Phase 1 Security Measures	phase1security_security_helmet, phase1security_security_csp, phase1security_security_xss_protection, phase1security_security_csrf_protection, phase1security_security_input_sanitiza…	1.00

5. Community Overview (Top 25)

The graph's 611 communities are ranked by size. Cohesion scores indicate internal connectivity: green ≥ 0.15 (well-structured), amber 0.08–0.15 (moderate), red < 0.08 (fragmented — consider refactoring).

ID	Name	Nodes	Cohesion	Top Members
5	Architecture & Audit Docs	48	0.05	Google AdSense Integration (Consent-gated), Anthropic Claude Haiku AI Integration, api/ai/route.ts (Claude Haiku Proxy),…
4	Dev Dependencies Config	47	0.04	devDependencies, eslint-config-prettier, husky, jest, lint-staged, prettier, ts-jest, ts-node
9	Turborepo Build Config	39	0.05	dependsOn, outputs, cache, cache, persistent, globalDependencies, globalEnv, dependsOn
2	SEO & Metadata Layer	35	0.09	generateMetadata(), getToolEmoji(), ToolPage(), BreadcrumbItem, Breadcrumbs(), BreadcrumbsProps, metadata, getRelatedToo…
18	Web App Dependencies	34	0.06	dependencies, bcryptjs, canvas, clsx, docx, fabric, html2canvas, ioredis
17	UI Package Dependencies	33	0.06	dependencies, class-variance-authority, clsx, @radix-ui/react-dialog, @radix-ui/react-dropdown-menu, @radix-ui/react-slo…
10	Medical Calculator Pages	32	0.06	breadcrumbSchema, FAQ_ITEMS, faqSchema, howToSchema, metadata, MISTAKES, RELATED_POSTS, _seo
16	Security Middleware (CSRF)	32	0.06	csrf, csrfErrorHandler(), csrfProtection, security, app, authRoutes, cookieParser, cors
1	Tool Client Components	31	0.04	CarbonResult, DIET_OPTIONS, FAQ, faqJsonLd, jsonLd, metadata, JsonLd(), JsonLdProps
6	UI Ad Components	31	0.07	AdAnchor(), AdAnchorProps, AdBanner(), AdBannerProps, AdInFeed(), AdInFeedProps, AdContext, AdContextValue
3	Calculator Input Types	28	0.05	BMIInput, BMIInput, LoanInput, LoanInput, MortgageInput, MortgageInput, ROIInput, ROIInput
23	API Server Dependencies	28	0.07	dependencies, @aws-sdk/client-s3, @aws-sdk/s3-request-presigner, bcryptjs, compression, cors, express, express-rate-limi…
7	API Auth & Billing Routes	27	0.06	ApiError, AppError, errorHandler(), AuthPayload, Request, requireAuth(), signToken(), affiliateRouter
13	API Middleware & Logger	27	0.08	rateLimit, apiKeyRateLimiter, authRateLimiter, calculatorRateLimiter, getRedisClient(), globalRateLimiter, makeRateLimit…
14	Claude / Skills Config	27	0.10	bool, DesignSystemGenerator, _detect_page_type(), format_ascii_box(), format_markdown(), format_master_md(), format_page…

25	Module Cluster 25	27	0.07	dependencies, @prisma/client, devDependencies, prisma, tsup, tsx, @types/node, typescript
26	Module Cluster 26	27	0.07	dependencies, bcrypt, cookie-parser, cors, csurf, dotenv, express, express-mongo-sanitize
22	Database Schema / Models	24	0.07	ForgotPasswordInput, forgotPasswordSchema, LoginInput, loginSchema, RegisterInput, registerSchema, ResetPasswordInput, r…
31	Module Cluster 31	24	0.11	applyPdfOptions(), ConvertFromFormat, convertPdfFormat(), ConvertToFormat, getMimeType(), GotenbergOptions, htmlToPdf(),…
12	Error Handling & Not Found	23	0.07	metadata, HomeSearchBar(), LazySection(), LazySectionProps, ToolGridSkeleton(), RelatedToolsProps, SearchBar(), CAT_COLO…
20	Calculators Page Layout	22	0.09	metadata, schemas, CALC_SUBCATEGORIES, CALC_TOOL_CONFIGS, CalcEntry, CalcSubcategory, FINANCIAL, FITNESS
41	Module Cluster 41	22	0.09	dependencies, zod, devDependencies, jest, ts-jest, @types/jest, typescript, main
28	Module Cluster 28	21	0.09	validateBody(), controller, dnsSchema, headers, networkRouter, socket, timeout, urlSchema
35	Module Cluster 35	21	0.12	classNames(), debounce(), deepClone(), formatCurrency(), formatDate(), generateId(), isEmpty(), sleep()
55	Module Cluster 55	21	0.10	devDependencies, autoprefixer, eslint, eslint-config-next, eslint-config-prettier, jest, jest-environment-jsdom, @playwr…

6. Architecture Findings

PDF Tool Pages — low cohesion (0.028)

The PDF tool community has 72 nodes but almost no internal edges (cohesion 0.028 — lowest in the graph). Each PDF tool page is an island with no shared layout abstraction beyond JsonLd(). Recommend extracting a PDFToolLayout wrapper component.

Tool Client Components — fragmented (0.040)

56 nodes, very few internal connections. This is effectively a 'junk drawer' community — components that share no common parent and drift independently. Consider grouping by feature subdirectory.

Duplicate AppError hierarchies

Communities 29, 30, and 173 all contain similar AppError/AuthenticationError/BadGatewayError trees from different source files. Three separate error class definitions means catch blocks may not match instances across package boundaries. Consolidate into packages/errors as the single source of truth.

packages/calculators-core — well-structured

137 files with cohesion scores of 0.39–0.45 for input/output schema clusters. The Zod schema/input/output pattern is consistent throughout. This is the best-structured package in the monorepo.

useTheme() in packages/ui is load-bearing

The graph detected useTheme() as a cross-community bridge with betweenness 0.022. SiteHeader pulls it from packages/ui/src/providers/theme.tsx directly. The custom hooks/useTheme.ts (now deleted) conflicted with this — the UI package's provider is correctly the sole theme authority.

JsonLd() is a single point of SEO failure

73 edges, betweenness 0.038. A regression here breaks structured data on every tool page simultaneously. This component has zero tests in the audit. Add at least a snapshot test for the JSON-LD output shape.

packages/db and lib/prisma.ts — no runtime conflict

Zero web-app files import @workspace/db. The web app exclusively uses lib/prisma.ts with the correct globalThis singleton guard. No duplicate connection pool risk in production.

2,925 isolated nodes — mostly noise

The majority are name/version/private fields parsed from package.json files. Not a code quality issue — AST noise from JSON parsing. Real isolated nodes worth investigating: generate_audit_pdf.py, design_system.py (skills directory — out of scope for production).

7. Knowledge Gaps

Nodes with 1 or fewer connections — possible missing edges, dead code, or under-documented components.

- **2925 isolated node(s):** `name`, `version`, `private`, `workspaces`, `dev` (+2920 more)
- **124 thin communities (<3 nodes) omitted from report** — run `graphify query` to explore isolated nodes.

8. Suggested Questions for Deep Exploration

These questions are uniquely answerable via the knowledge graph — run `graphify query "<question>"` in the project root for a traced answer.

- Q1: Why does JsonLd() bridge 60+ tool page communities to the SEO layer, and what breaks if it fails?
- Q2: Should PDF Tool Pages be split into smaller, more focused modules? (cohesion 0.028)
- Q3: Should Tool Client Components be split into smaller, more focused modules? (cohesion 0.040)
- Q4: What connects the isolated documentation nodes to the rest of the system?
- Q5: Why does SiteHeader() bridge the theme provider to the navigation cluster?
- Q6: How does authentication flow from the register/login endpoints to the Prisma client?
- Q7: What is the shortest path from the AdSense integration to the tool page rendering pipeline?

9. Remediation Summary — All 9 Phases

The following issues were identified and fixed prior to this audit report being generated.

Phase	Changes
Phase 1 — Security	Hashed passwords in register route using bcrypt (cost 12). Plaintext storage eliminated.
Phase 2 — Brand & SEO	Replaced all 'Toolverse' references with 'RoughTools' across seo-config.ts, structured-data.ts, 9 sitemap files, nginx.conf, and PDF tool canonicals. Fixed sitemap PDF category mapping (document-tools → pdf). Updated tools-registry comment from 194 → ~415 tools.
Phase 3 — Broken Pages	About page: replaced legacy Navbar with SiteHeader. Contact form: replaced alert() with Resend API route + proper success/error states. Pricing page: added href to Pro and Team CTAs. Dashboard: added 8s AbortController timeout, skeleton loading, error/retry state.
Phase 4 — Typography	Fixed Tailwind fontFamily config to reference CSS variables (var(--font-display) etc.) matching actual loaded fonts. ToolCard description: leading-snug → leading-normal.
Phase 5 — Theme Conflicts	Deleted hooks/useTheme.ts (conflicted with next-themes ThemeProvider). Deleted unused HomeDarkMode.tsx.
Phase 6 — Component Cleanup	Deleted components/ui/ToolCard.tsx (dead, only imported by deleted DocumentsSection). Deleted components/layout/ directory (zero importers post-Phase 3). Deleted components/sections/ directory (zero external importers). Deleted components/tools/MortgageCalculator.tsx (orphan — mortgage page uses local client).
Phase 7 — AdSense & Analytics	CopyButton: removed success toast (icon morph only, keeping failure toast). ToolAdUnit sidebar: added position:sticky top:72px. WebVitals: added GA_MEASUREMENT_ID guard + dev warning. Documented NEXT_PUBLIC_GA_MEASUREMENT_ID and NEXT_PUBLIC_CLARITY_PROJECT_ID in .env.example.
Phase 8 — API & Config	Deleted stray src/app/next.config.js. Fixed images.remotePatterns (replaced **.example.com with real domains). AI model claude-haiku-4-5-20251001 confirmed current. Prisma duplicate confirmed non-issue (web app uses lib/prisma.ts only).
Phase 9 — Infrastructure	Nginx: removed X-Frame-Options SAMEORIGIN (conflicts with Next.js DENY). Dockerfile: added 11 missing package COPY statements for pnpm workspace. robots.ts: allowing GPTBot/Claude-Web/PerplexityBot confirmed intentional.